

- 1 Which of the following is not a reasonable estimate?
- A Sound of frequency in the order of 10^9 Hz is audible to human beings.
 - B Light of wavelength in the order of 10^{-7} m is visible to human beings.
 - C The volume of an average-sized orange is in the order of 10^{-4} m³.
 - D The speed of beta particles emitted from a nucleus is in the order of 10^7 m s⁻¹.

Ans: A

Options B and D are typical values that candidates should be familiar with. Using $V = \frac{4}{3}\pi r^3$,

Option C works out for an orange having radius of about 5 cm. The frequency of sound (or even ultrasound) cannot hit GHz range.

- 2 The kinetic energy E of a rotating solid sphere is given by the following expression